Statistics and Data Analysis

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Statistics and Data Analysis (A00914)

Short Title: Statistics and Data Analysis
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author KGRENNAN
Credits 10 **Level:** Postgraduate

Description of Module / Aims

The module aims to develop the learner's competence in the application of modern techniques of data analysis in the context of analytical method development. The learner will be enabled to apply statistical analysis processes to data sets and interpret the results in order to quantify data quality and compare data sets. These data sets will deal with real world applications from the pharmaceutical and food industries. The learner will also develop competence in a range of process control tools and minimisation strategies. Continuous improvement and data-driven decisions will also be emphasised throughout the course of the programme.

Programmes

`STAT-0052`	Certificate in Statistics and Data Analysis (`WD_STATD_MA`)	5 / 0 / M
`STAT-0052`	MSc in Analytical Science with Quality Management (`WD_SANSC_R`)	1 / 0 / E
`STAT-0052`	PG Dip in Analytical Science with Quality Management (`WD_SANSC_G`)	1 / 0 / E

Indicative Content

- Descriptive statistics: graphical methods of display, measures of centre and spread
- Precision and error measurement: relative and fixed bias, random errors, propagation of errors and accuracy, comparing and combining standard deviations and means of data sets, limit of detection
- Sampling distributions: point estimation, sampling from continuous and discrete distributions
- Statistical inference: inference for single, two or more samples
- Analysis of data through case studies in `real world' applications in the pharmaceutical and food industries
- Statistical process control: multivariate statistical process control, process monitoring and control strategies, process capability indices in statistics, Shewhart control charts, CUSUM charts, moving average and moving range charts
- Sampling plans and the statistical analysis of the quality of batches

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Assess analytical data using appropriate descriptive statistics tests.
2. Assess data sets based on applied inter-comparison significance tests.
3. Assess analytical data using statistical software packages.
4. Assemble and integrate process capability indices as a statistical measure of process capability.
5. Justify the usefulness of Shewhart control charts, CUSUM charts and other control charts to control a process.
6. Evaluate sampling plans and statistical analysis of quality in product batches.

Learning and Teaching Methods

- Lectures.

- Tutorials.
- IT practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Tutorial/Problem Sheets	30%	1,2
Practical	30%	3
In-Class Assessment	40%	4,5,6

Assessment Criteria

<40%: The learner will demonstrate little or no understanding of the fundamentals of the learning outcomes. Will display a lack of competence with regards to associated data analysis skills and statistics.

40%-59%: The learner will demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regards to associated data analysis skills and statistics.

60%-69%: As well as the above, the learner will demonstrate reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to associated data analysis skills and statistics.

70%-100%: All of the above to excellent levels. In addition, the learner will demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to associated data analysis skills and statistics.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Tutorial		12
Practical		12
Independent Learning		222

Essential Material

- "Statistics & Data Analysis module Moodle page." http://moodle.wit.ie
- Montgomery, D.C. _Introduction to Statistical Quality Control_. 7th ed. New York: Wiley, 2013.

Supplementary Material

- "Science Direct." http://www.sciencedirect.com

Requested Resources

- Room Type: Computer Lab test